

31/07/26

CO2 AT 2

1. data preprocessing in retail analytics

A. Handling missing values

customer	Missing column
C2	Age
C3	Monthly - Spend
C5	Purchase - Frequency

Technique 1: Median

* Age values = 23, 31, 105, 27, 40, 36, 42

* Median Age = 36

* Monthly spend values = 20, 35, 100, 25, 45, 300

* Median spend = 38

* purchase frequency = 5, 8, 10, 50, 12, 9, 45

* Median = 10

Imputed values;

* C2 Age = 36

* C3 Monthly spend = 38

* C5 purchase frequency = 10

Technique: 2; K-NN Imputation

Find nearest similar records based on available attributes.

Eg:

* C2 Age $\approx 34-36$

* C3 Monthly spend ≈ 40

* C5 purchase frequency $\approx 9-10$

Justification;

Median is robust to outliers while K-NN uses neighbours records for more accurate estimation.

B outlier detection

extreme values;

* Age = 105

* Monthly spend = 500

These have very high Z-scores & can be treated as outliers.

Treatment;

* cap age to upper acceptable limit (say

* Cap Monthly spend to ≈ 100 or some
the record of necessary.

Reason ;

without reduce model accuracy.

c, categorical data standardization

original gender values

* M

* male

* male

* Female

* Female

standardized

original	standard
M	male
male	male
male	male
MALE	male
F	Female
Female	Female
Female	Female
FEMALE	Female

2. Covariance & data Reduction in e-commerce Analytics

A, Covariance Analysis

Income & purchase value both increase together.

Therefore,

covariance = positive

Interpretation;

* High income customer usually purchase more.

* positive covariance indicates a direct relationship.

covariance helps identify relationship between variables before Model building.

B, Feature selection

Highly correlated features

* Income

* purchase value

* Average Balance

Selected Features

* Age

* Income

* Credit score

* EMI

Removed

* Average balance

* EMI

Reason :

They provide similar information

→ Increase redundancy

c, Data Transformation

Apply standardization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Benefits :

* Remove scale differences

* Improves ML performance

* prevents large value from dominating

3. Data Transformation & Standardization

A. Equal width binning

study hours

Minimum = 2

Maximum = 11

Range = 9

NO of bins = 3

Bin width = 3

Bins

* 2-5 → Low

* 5-8 → Medium

* 8-11 → High

test score

Min = 40

Max = 100

Range = 60

width = 20

Bins

* 40-60

* 60-80

* 80-100

B, Equal Frequency Binning
Each bin contains approximately equal
records.

study hours

Bin 1

2, 3, 4

Bin 2

5, 6, 7

Bin 3

8, 9, 10, 11

C, Concept Hierarchy

study hours

* low

* med

* high

Test score

* poor

* average

* excellent

4. normalization

dataset

200, 300, 400, 600, 1000

A, Min-Max normalization (0-1)

Formula

$$\frac{x - \min}{\max - \min}$$

Results

value	normalized
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

B, z-score normalization

mean;

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5}$$

$$= 500$$

standard deviation ≈ 282.84

Results

value	Z-score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

5. online shopping behavior analysis

unit duration

$$\frac{11-8}{2011} = 1$$

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

A. Equal frequency binning

Bin 1

15, 30, 45

Bin 2

60, 120, 300

Bin 3

600, 900, 1800, 3600

analysis

- * short visits grouped together
- * long visits grouped together
- * useful for customer segmentation

B, Z-score normalization

Mean

= 747 seconds

standard deviation $\approx$ 1102 seconds

Formula

$$Z = \frac{x - \mu}{1102}$$

Example

$$* 15 \text{ sec} = -0.66$$

$$* 3600 \text{ sec} = 2.59$$

Discussion

extreme values remain visible
but become comparable after scaling.

C, min-max normalization

Formula

$$\frac{x-15}{3600-15}$$

Sample Results

duration	normalized
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison :

* Min-Max normalization : scale value between 0 & 1 but is sensitive to outliers.

* Z-score normalization : centers data around the mean & is better for machine learning when outliers are present.